Exact and stable games need not be egalitarian stable

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Abstract

Dietzenbacher (Games and Economic Behavior 127, 2021) asked whether every exact game and every stable-core game is egalitarian stable, that is, whether the procedural egalitarian solution always selects a core allocation on these classes; by known results this holds for exact games with at most four players and stable games with at most five. We answer both halves of the question in the negative. For exactness: there is a five-player exact game — monotone, superadditive, with grand-coalition worth 4 — whose egalitarian claims all equal 1, so the claims sum to $5 > 4$ and the procedural egalitarian solution leaves the core. A padding construction yields exact non–egalitarian-stable games for every $n \geq 5$, so exactness implies egalitarian stability precisely up to four players. For stability: there is a six-player exact game with a von Neumann–Morgenstern stable core that is not egalitarian stable, and the threshold $n = 6$ is again sharp. Both witnesses arise from one device: the pointwise minimum of finitely many efficient vectors is always an exact game. The stable witness has a one-dimensional core, and for such segment-core games we prove that core stability admits a finite certificate — an arrangement-vertex enumeration — which we discharge in exact rational arithmetic. We further isolate the mechanism (a strict over-cover of the claims) and explain why random sampling is unlikely to find such games.

Keywords: cooperative games; core; exact games; egalitarian stability; procedural egalitarian solution.

1 Introduction

Egalitarianism and core stability interact delicately in transferable-utility (TU) games. The procedural egalitarian solution (Dietzenbacher et al. 2017) extends the constrained egalitarian solution of Dutta and Ray (1989) from convex games to all TU games through an iterative procedure of intercoalitional equal division; Dietzenbacher (2021) showed that this solution selects a core allocation exactly on the class of *egalitarian stable* games, proved that all games with a large core (Sharkey 1982) are egalitarian stable, and—noting that by Biswas et al. (1999) every exact game (Schmeidler 1972) with at most four players has a large core—posed the following question: "Whether all exact games and all stable core games with an arbitrary number of players are egalitarian stable is an interesting open question for future research."

*A machine-readable certificate supplement with a standalone verifier accompanies this paper; see the verification section for methodology and the disclosure of AI assistance.

This paper answers both halves of the question in the negative, each at the smallest possible size.

- **A five-player counterexample (Section 4).** There is an exact, monotone, superadditive five-player game with $v(N) = 4$ whose egalitarian claims are $(1, 1, 1, 1, 1)$: every player's claim is locked at 1 by the procedure, yet the grand coalition can only pay 4. The game is generated by a device of independent interest: the pointwise minimum of finitely many efficient payoff vectors is always an exact game (Lemma 2.1). Two further, structurally different witnesses found by independent searches are reported alongside.

- **A sharp threshold (Section 5).** A padding operation produces from any witness an exact non–egalitarian-stable game with any number $n \geq 5$ of players, with a complete proof for null padding. Combined with the positive result for $n \leq 4$, exactness implies egalitarian stability if and only if $n \leq 4$.

- **A six-player stable counterexample (Section 7).** There is an exact six-player game with a stable core that is not egalitarian stable; since largeness and core stability coincide for $n \leq 5$ (Estévez-Fernández 2012) and large-core games are egalitarian stable, the threshold $n = 6$ is sharp. Its core is a one-dimensional segment, and we prove a finite stability certificate for segment-core games (Proposition 7.4): the undominated imputations form a finite union of polytopes whose vertices lie in an explicit plane arrangement, so enumerating the arrangement's vertices decides stability. The certificate is discharged computationally in exact rational arithmetic.

- **Mechanism and method (Section 6).** A game is egalitarian stable if and only if its claims vector lies in the core (Lemma 3.1); failure requires a strict over-cover of the claims, which in all known witnesses over-counts a single player. We also explain why the question resisted sampling: exactified random games almost never leave the large-core region, where stability is guaranteed; the witnesses live on the thin-core frontier that the minimum-of-points generator produces directly.

All claims are certified in exact rational arithmetic; the procedure implementations were validated against the worked examples of Dietzenbacher (2021), and every witness was verified through several independently written implementations (Section 8).

2 Preliminaries and the egalitarian procedure

A *transferable-utility (TU) game* is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v\colon 2^N \to \mathbb{R}$, $v(\emptyset) = 0$; we write v for the game and $x(S) = \sum_{i \in S} x_i$ for $x \in \mathbb{R}^N$. The *core* is

$$\mathrm{Core}(v) = \big\{ x \in \mathbb{R}^N \ : \ x(N) = v(N),\ x(S) \geq v(S)\ \forall S \subseteq N \big\},$$

and v is *balanced* when $\mathrm{Core}(v) \neq \emptyset$. The game is *superadditive* if $v(S \cup T) \geq v(S) + v(T)$ for disjoint S, T, *monotone* if $v(S) \leq v(T)$ for $S \subseteq T$, and *convex* if $v(S \cup T) + v(S \cap T) \geq v(S) + v(T)$ for all S, T. Following Schmeidler (1972), v is *exact* if for every $S \subseteq N$ there is a core element attaining $v(S)$, equivalently

$$\min_{x \in \mathrm{Core}(v)} x(S) = v(S) \qquad \text{for all } S \subseteq N.$$

The core is *large* (Sharkey 1982) if every $x \in \mathbb{R}^N$ with $x(S) \geq v(S)$ for all S dominates a core element from below. One has convex $\subseteq$ large core $\subseteq$ exact $\subseteq$ totally balanced, with all inclusions strict for $n \geq 5$.

Exactness admits a convenient finite generator, which we use to build and certify witnesses. It is the finite specialization of Schmeidler's lower-envelope description of exact games (an exact game is the lower envelope of its own core); we record it explicitly.

Lemma 2.1 (finite generators of exact games)**.** *Let $P \subseteq \mathbb{R}^N$ be a nonempty finite set of vectors sharing a common sum τ, and define $v(S) = \min_{x \in P} x(S)$. Then $v(N) = \tau$, every $x \in P$ lies in* $\mathrm{Core}(v)$*, and v is exact.*

Proof. Every $x \in P$ has $x(N) = \tau = v(N)$ and $x(S) \geq \min_{y \in P} y(S) = v(S)$ for all S, so $x \in \mathrm{Core}(v)$; in particular $\mathrm{Core}(v) \neq \emptyset$. Fix S and pick $x^* \in \arg\min_{x \in P} x(S)$. Then $x^* \in \mathrm{Core}(v)$ and $x^*(S) = v(S)$, so $\min_{x \in \mathrm{Core}(v)} x(S) \leq v(S)$; the reverse inequality holds because every core element satisfies $x(S) \geq v(S)$. Hence $\min_{\mathrm{Core}(v)} x(S) = v(S)$ for all S. □

The egalitarian procedure. We recall the procedure underlying the *procedural egalitarian solution* (Dietzenbacher et al. 2017; Dietzenbacher 2021). Set $P^{v,0} = \emptyset$. For $k \geq 1$ and $S \neq \emptyset$ define $\chi^{v,k}(S) \in \mathbb{R}^S$ by

$$\chi_i^{v,k}(S) = \begin{cases} \gamma_i^{v,k-1}, & i \in S \cap P^{v,k-1}, \\ \dfrac{v(S) - \sum_{j \in S \cap P^{v,k-1}} \gamma_j^{v,k-1}}{|S \setminus P^{v,k-1}|}, & i \in S \setminus P^{v,k-1}, \end{cases}$$

i.e. already-fixed members keep their payoff and the residual worth of S is split equally among the rest. The *egalitarian admissible* collection of round k is

$$A^{v,k} = \Big\{ S \;:\; \textstyle\sum_{i \in S} \chi_i^{v,k}(S) = v(S) \;\text{ and }\; \forall i \in S\ \forall T \ni i :\ \chi_i^{v,k}(T) \leq \chi_i^{v,k}(S) \Big\},$$

so S is admissible when it is efficient for $\chi^{v,k}$ and gives each of its members their pointwise-largest $\chi^{v,k}$-value. Put $P^{v,k} = \bigcup_{S \in A^{v,k}} S$ and $\gamma_i^{v,k} = \chi_i^{v,k}(S)$ for $i \in P^{v,k}$, $S \in A^{v,k} \ni i$. The procedure terminates at $n_v = \min\{k : P^{v,k} = N\}$; the *egalitarian claims* vector is $\widehat{\gamma}^{\,v} = \gamma^{v,n_v}$ and $\widehat{A}^v = A^{v,n_v}$.

The claims vector is always an *aspiration* (Bennett 1983): $\widehat{\gamma}^{\,v}(S) \geq v(S)$ for all S, and for each $i \in N$ there is $S \ni i$ with $\widehat{\gamma}^{\,v}(S) = v(S)$ and $\widehat{\gamma}_i^v \leq \widehat{\gamma}_j^v$ for all $j \in S$. A game is *egalitarian stable* (ES) if $N \in \widehat{A}^v$; equivalently, the procedural egalitarian solution then selects from the core. Convex and large-core games are ES (Dietzenbacher 2021). Since exact games with $n \leq 4$ have large cores (Biswas et al. 1999), they are ES; whether every exact game is ES was left open (Dietzenbacher 2021, Sec. 4). We resolve it negatively.

3 ES equals membership of the claims vector in the core

The following characterization reduces egalitarian stability of any balanced game to a single membership test and organizes everything below. Write $g(S) = \widehat{\gamma}^{\,v}(S) - v(S) \geq 0$ for the aspiration gap. A collection of weights $\lambda = (\lambda_S)_{S \subsetneq N}$, $\lambda_S \geq 0$, is a *fractional cover* if $\sum_{S \ni i} \lambda_S \geq 1$ for every $i \in N$.

Lemma 3.1. *Let v be balanced with egalitarian claims $\widehat{\gamma} = \widehat{\gamma}^{\,v}$. The following are equivalent:*

(i) *v is egalitarian stable;*

(ii) $\widehat{\gamma} \in \mathrm{Core}(v)$*;*

(iii) $\sum_{i \in N} \widehat{\gamma}_i = v(N)$*;*

(iv) *$\widehat{\gamma}$ dominates a core element from below;*

(v) $g(N) \le \min\{\sum_{S \subsetneq N} \lambda_S \, g(S) \ : \ \lambda$ *a fractional cover*$\}$.

Proof. $\widehat{\gamma}$ is an aspiration, so $\widehat{\gamma}(S) \ge v(S)$ for all S and $\sum_{i \in N} \widehat{\gamma}_i \ge v(N)$ always.

(i)⇒(iii). If v is ES then $N \in \widehat{A}^v$, and the efficiency requirement in the definition of A^{v,n_v} gives $\sum_{i \in N} \chi_i^{v,n_v}(N) = v(N)$; at the terminal round every player is fixed, so $\chi_i^{v,n_v}(N) = \widehat{\gamma}_i$ and $\sum_i \widehat{\gamma}_i = v(N)$.

(iii)⇒(ii). With $\sum_i \widehat{\gamma}_i = v(N)$ and $\widehat{\gamma}(S) \ge v(S)$ for all S, $\widehat{\gamma}$ is efficient and satisfies every core inequality, i.e. $\widehat{\gamma} \in \mathrm{Core}(v)$.

(ii)⇒(iv) is trivial. (iv)⇒(i): let $y \in \mathrm{Core}(v)$ with $y \le \widehat{\gamma}$. For every $S \in \widehat{A}^v$, $y(S) \le \widehat{\gamma}(S) = v(S) \le y(S)$, so $y(S) = \widehat{\gamma}(S)$ and, as $y \le \widehat{\gamma}$, $y_i = \widehat{\gamma}_i$ for $i \in S$. The admissible coalitions cover N, so $y = \widehat{\gamma} \in \mathrm{Core}(v)$; at the terminal round N is then admissible, i.e. v is ES.

(iv)⇔(v). With $y = \widehat{\gamma} - t$, domination of a core element is feasibility of $t \ge 0$, $t(N) = g(N)$, $t(S) \le g(S)$ $(S \subsetneq N)$; this holds iff $\max\{t(N) : t \ge 0, \ t(S) \le g(S)\} \ge g(N)$, and LP duality identifies the maximum with $\min\{\sum_S \lambda_S g(S) : \lambda$ a fractional cover$\}$. □

Remark 3.2. *By Bondareva–Shapley, every fractional* partition *(one with $\sum_{S \ni i} \lambda_S = 1$) obeys $\sum_S \lambda_S v(S) \le v(N)$, and hence $\sum_S \lambda_S g(S) \ge g(N)$. A failure of* (v) *must therefore be certified by a* strict over-cover*, one in which some player is covered with total weight exceeding* 1*. This is the analytic signature of the witnesses below: the egalitarian claims inflate precisely because one pivotal player is over-covered by the tight coalitions that fix the claims.*

4 A five-player exact game that is not egalitarian stable

Theorem 4.1. *There is an exact five-player game that is not egalitarian stable. Consequently the implication "exact ⇒ ES," valid for $n \le 4$, fails at $n = 5$.*

Proof. Let $N = \{1, 2, 3, 4, 5\}$ and take the two allocations

$$x^1 = (0, 0, 1, 1, 2), \qquad x^2 = (1, 1, 0, 2, 0),$$

both with total 4, and set $v(S) = \min\{x^1(S), x^2(S)\}$ for all S. By Lemma 2.1, v is exact, with $x^1, x^2 \in \mathrm{Core}(v)$ and $v(N) = 4$; the game is monotone and superadditive but not convex, and its full coalition function is listed in the supplement. In fact x^1 and x^2 are the *only* vertices of $\mathrm{Core}(v)$, so the core is the one-dimensional segment $\mathrm{conv}\{x^1, x^2\}$—the thinnest nondegenerate core, and exactly the regime in which non-egalitarian-stability concentrates (Remark 6.3).

The procedure. The maximum average worth is $\max_S v(S)/|S| = 1$, attained exactly on

$$\widehat{A}^v = \big\{\{4\}, \{3, 4\}, \{4, 5\}, \{1, 4, 5\}, \{2, 4, 5\}\big\},$$

Table 1: Egalitarian procedure for the primary witness (single round, cf. Dietzenbacher 2021, Ex. 2). Entries are $\chi_i^{v,1}(S)$ over $i = 1, \ldots, 5$, "·" for $i \notin S$; the last column is N.

S	$\{4\}$	$\{3,4\}$	$\{4,5\}$	$\{1,4,5\}$	$\{2,4,5\}$	N
$v(S)$	1	2	2	3	3	4
$\chi^{v,1}(S)$	$(\cdot,\cdot,\cdot,1,\cdot)$	$(\cdot,\cdot,1,1,\cdot)$	$(\cdot,\cdot,\cdot,1,1)$	$(1,\cdot,\cdot,1,1)$	$(\cdot,1,\cdot,1,1)$	$(\frac{4}{5},\frac{4}{5},\frac{4}{5},\frac{4}{5},\frac{4}{5})$

each of which routes through the pivot player 4 (see the single-round trace, Table 1). Every player attains its own maximum average, 1, on one of these coalitions, so all five are egalitarian admissible in the first round; thus $n_v = 1$ and

$$\widehat{\gamma}^{\,v} = (1,1,1,1,1), \qquad \sum_{i \in N} \widehat{\gamma}_i = 5 > 4 = v(N).$$

By Lemma 3.1 the game is not egalitarian stable: the grand coalition would give each player only $v(N)/5 = \frac{4}{5} < 1$, so $N \notin \widehat{A}^v$. □

Here the egalitarian claims are *symmetric*, yet the game is not ES: what inflates them is not an unequal claim but the fact that every tight coalition is routed through the single pivot 4 (the over-cover of Remark 3.2). The inclusion-maximal admissible coalitions $\{3,4\}, \{1,4,5\}, \{2,4,5\}$ have $\{4\}$ as their only common member, so the strong egalitarian claimants are $D^v = \{4\}$ and the procedural egalitarian solution equalizes the rest down to $\lambda = \frac{3}{4}$,

$$PES(v) = \left(\tfrac{3}{4}, \tfrac{3}{4}, \tfrac{3}{4}, 1, \tfrac{3}{4}\right),$$

which leaves the core, violating $\{3,4\}$ and $\{4,5\}$ by $\frac{1}{4}$ and $\{1,4,5\}$ and $\{2,4,5\}$ by $\frac{1}{2}$. A core allocation attaining $v(S)$ for each of the 30 proper coalitions (the exactness certificate) is provided in the supplement.

Remark 4.2 (a minimal-mass variant)**.** *The grand-coalition worth $v(N) = 4$ is minimal over two-point generators, and the five tight coalitions that drive the procedure keep their worths under coalitionwise reduction. Greedily lowering the non-tight worths yields a variant of total worth-mass* 41 *(from* 46*) that is again exact, monotone and superadditive, has the identical procedure trace (Table 1), the same claims, and the same* PES *with the same four core violations, and is locally minimal—no single worth admits a further decrease preserving these properties (supplement). Its core has seven vertices, so the two-point generator and the segment core are particular to the reference game above.*

Remark 4.3. *By Lemma 3.1 the witness is exact but* not *large-core, so it lies in the gap between exact and large-core games—empty for $n \leq 4$, first opening at $n = 5$ (Biswas et al. 1999). Two further, independently located witnesses confirm the phenomenon and its two modes: an exact game with worths at most* 6 *and an* asymmetric *pivot claim, $\widehat{\gamma} = (1,1,3,1,1)$, where pivot player* 3 *carries claim* 3 *and $\sum_i \widehat{\gamma}_i = 7 > 6 = v(N)$; and an exact game whose claims $\frac{1}{2}(2,7,3,7,2)$ sum to $\frac{21}{2} > 10 = v(N)$. All three were mutually cross-verified across independent implementations; their data are in the supplement.*

5 An infinite family: exact non-ES games for every $n \geq 5$

We upgrade Theorem 4.1 to a family covering all $n \geq 5$ by a padding operation that provably preserves both exactness and non-ES.

Definition 5.1. *Let v be a game on N and $c \geq 0$. The* c-padding *of v is the game v^+ on $N \cup \{p\}$ ($p \notin N$) with $v^+(S) = v(S)$ for $S \subseteq N$ and $v^+(S \cup \{p\}) = v(S) + c$. For $c = 0$, p is a* null *player; for $c \geq 1$, p is essential.*

Lemma 5.2 (padding preserves exactness)**.** *If v is exact then so is every c-padding v^+ ($c \geq 0$).*

Proof. Let $x \in \mathrm{Core}(v^+)$. From $x(N) \geq v(N)$, $x_p \geq v^+(\{p\}) = c$ and $x(N) + x_p = v^+(N) = v(N) + c$ we get $x(N) = v(N)$ and $x_p = c$; conversely any (y, c) with $y \in \mathrm{Core}(v)$ lies in $\mathrm{Core}(v^+)$. Hence $\mathrm{Core}(v^+) = \mathrm{Core}(v) \times \{c\}$, and $\min_{\mathrm{Core}(v^+)} x(S) = \min_{\mathrm{Core}(v)} y(S \setminus \{p\}) + c\,[p \in S] = v(S \setminus \{p\}) + c\,[p \in S] = v^+(S)$ for every S. □

Lemma 5.3 (null players are inert)**.** *Let v be nonnegative and v^+ its 0-padding by a null player p. Then the egalitarian procedure on v^+ fixes each $i \in N$ in the same round and at the same value as on v, and fixes p last at value 0; in particular $\widehat{\gamma}^{v^+} = (\widehat{\gamma}^{v}, 0)$.*

Proof. By induction on k: $P^{v^+,k} \setminus \{p\} = P^{v,k}$, $\gamma^{v^+,k}$ agrees with $\gamma^{v,k}$ on $P^{v,k}$, and $p \notin P^{v^+,k}$ while $P^{v,k} \neq N$. The base is trivial. Assume it for $k-1$, with $Q = P^{v,k-1}$, claims γ, and p free. For $S \subseteq N$ the data defining $\chi^{v^+,k}(S)$ and $\chi^{v,k}(S)$ coincide, so they are equal. For $R = S \cup \{p\}$ with $f = \gamma(Q \cap S)$: if $S \setminus Q \neq \emptyset$, each free member of R gets $\sigma = (v(S) - f)/(|S \setminus Q| + 1)$, whereas $\chi_i^{v^+,k}(S) = (v(S) - f)/|S \setminus Q|$ for $i \in S \setminus Q$; if $v(S) - f \geq 0$ then $\sigma \leq \chi_i^{v^+,k}(S)$, and if $v(S) - f < 0$ then $\sigma < 0 \leq v(\{i\}) \leq \max_{T \ni i} \chi_i^{v^+,k}(T)$ (nonnegativity, with $\{i\}$ available). Either way no original free member attains a new maximum at R, so no such R is admissible in round k; the only admissible coalitions containing p have all original members already in Q, and they fix p alone. Hence the newly fixed original players match those of v, proving the step. At termination $\widehat{\gamma}_p^{v^+} = \max_{S \subseteq N}(v(S) - \widehat{\gamma}^{v}(S)) \leq 0$ (aspiration) and $\geq v^+(\{p\}) = 0$, so $= 0$. □

Theorem 5.4. *For every $n \geq 5$ there is an exact game on n players that is not egalitarian stable. Concretely, padding the primary witness of Theorem 4.1 with $n - 5$ null players yields such a game.*

Proof. The witness v ($n = 5$) is exact, nonnegative, and non-ES with $\sum_i \widehat{\gamma}_i^{v} = 5 > 4 = v(N)$. Adjoin a null player: by Lemma 5.2 the padding v^+ is exact, and by Lemma 5.3 $\widehat{\gamma}^{v^+} = (\widehat{\gamma}^{v}, 0)$, so $\sum_i \widehat{\gamma}_i^{v^+} = 5 > 4 = v^+(N)$; by Lemma 3.1 v^+ is not ES. Nonnegativity and both properties are preserved, so iterating $n - 5$ times gives an exact non-ES game for each $n \geq 5$. □

Null players are convenient for the proof but degenerate. Padding with *essential* players ($c \geq 1$) gives non-degenerate examples with the same inflation; we verified (supplement) that for $c = 1$ the padded player receives claim c, all other claims are unchanged, and the games remain exact and non-ES through $n = 9$, with every player essential.

Corollary 5.5 (sharp threshold)**.** *Every exact game with $n \leq 4$ is egalitarian stable, and for every $n \geq 5$ there is an exact game that is not. The threshold $n = 5$ coincides with the onset of the exact/large-core gap.*

Proof. For $n \leq 4$, exact games have large cores (Biswas et al. 1999) and large-core games are ES (Dietzenbacher 2021). For $n \geq 5$, Theorem 5.4. □

6 The mechanism, and why the question stayed open

Lemma 3.1 and Remark 3.2 isolate the combinatorial cause of non-ES. We record it as a sufficient condition covering both observed modes, then explain why a naive symmetric family cannot realize it and why the phenomenon eluded random sampling.

Proposition 6.1 (pivot over-cover)**.** *Let v be exact, and suppose there is a player h and tight coalitions $T_1, \dots, T_r \ni h$ (each $\widehat{\gamma}(T_j) = v(T_j)$) that cover N, are each fixed by the egalitarian procedure, and each spend h's full claim $\widehat{\gamma}_h$. Then $\{T_j\}$ is a strict over-cover of N at h, whence $\sum_i \widehat{\gamma}_i > v(N)$ and v is not egalitarian stable.*

Proof. Each T_j is tight, so unit weights on the T_j give a fractional cover with $\sum_j g(T_j) = 0$; as every T_j contains h, the cover assigns h total weight ≥ 2, i.e. it is strict, and Lemma 3.1 (v) fails once $g(N) > 0$. Equivalently, summing $\widehat{\gamma}(T_j) = v(T_j)$ double-counts $\widehat{\gamma}_h$, forcing $\sum_i \widehat{\gamma}_i > v(N)$. □

Proposition 6.1 unifies the two modes seen in Remark 4.3. In the *asymmetric* mode (the hub witness) the pivot $h = 3$ additionally carries an inflated claim $\widehat{\gamma}_h = 3$. In the *symmetric* mode (the primary witness) all claims equal 1, yet the five tight coalitions still route through the single pivot $h = 4$; the symmetric mode is exactly the special case of equal claims. In both, exactness survives because "absorber" players let the core attain the pivot's small-coalition worths (for the primary witness, x^1 and x^2 attain the relevant worths from opposite sides): the tight coalitions certifying the claims are pairwise incompatible—all require the pivot—yet each individual worth remains core-attainable.

Remark 6.2 (symmetric scaling collapses to ES)**.** *A tempting symmetric scaling—a hub h with $v(\{h\}) = m$, one absorber D, and k leaves, with $v(S) = m + 2$ for $h, D \in S$ and ≥ 1 leaf, $v(\{h, D\}) = m + 1$, $v(N) = m + 3$—is non-ES for all $k \geq 3$ (claims $(m, 1, \dots, 1)$, sum $m+1+k > m+3$) but is* not exact*: $\min_{\text{Core}} x(\{h, D\}) = m+2 > m+1$, since $x(\{h, D, \text{leaf}\}) \geq m + 2$ with $x(N) = m + 3$ pins every leaf to 0. Raising $v(\{h, D\})$ to its core-minimum—the exactification forced by exactness—hands the surplus to the absorber (claim $\frac{3}{2}$, leaves $\frac{1}{2}$) and the game becomes ES; its $n = 4$ member is exact* and *ES, as Corollary 5.5 requires. A genuine exact non-ES game must therefore be asymmetric—several satellites and absorbers, as in the witnesses—which is why padding an existing witness, not scaling a symmetric template, is the robust route to Theorem 5.4.*

Remark 6.3 (why the question stayed open)**.** *Non-ES exact games are not pathological once one looks in the right place, but they are nearly invisible to the natural sampling scheme. Exactifying random balanced games essentially never leaves the large-core class (in our sampling,* 3 *of* 499 *exactified games were even non-large-core, and a random-aspiration probe detected non-largeness in none), so such a search almost never produces a non-ES game. By contrast, drawing exact games as lower envelopes of a few points (Lemma 2.1) lands on the thin-core frontier where the phenomenon concentrates: among* 18,507 *point-generated exact five-player games,* 493 *(*2.66%*) were not egalitarian stable. Within these, the symmetric-claims mode of the reference witness is itself rare—only* 2 *of the* 493 *have all claims equal—and the claim inflation $\sum_i \widehat{\gamma}_i - v(N)$ concentrates at the values* 1 *and* $\frac{1}{2}$*. The offending aspiration is always the egalitarian claims vector itself, which generic aspiration sampling misses; this mismatch between the natural sampling scheme and the location of the counterexamples is, we suspect, why the question remained open.*

7 Stable games need not be egalitarian stable

We now resolve the second half of the question. Recall that an *imputation* is an efficient, individually rational allocation, and that an allocation x *dominates* an imputation y via S if $x_i > y_i$ for all $i \in S$ and $x(S) \le v(S)$ (von Neumann and Morgenstern 1944). The core, always internally stable, is a *(von Neumann–Morgenstern) stable set* precisely when it is also externally stable: every imputation outside the core is dominated by a core element. We then say the game has a *stable core*. Large-core games have stable cores (Sharkey 1982), and for games with at most five players largeness and core stability are equivalent (Estévez-Fernández 2012, Sec. 5); since large-core games are egalitarian stable (Dietzenbacher 2021), every stable game with $n \le 5$ is egalitarian stable. Dietzenbacher (2021) asked whether this persists for all n. It does not:

Theorem 7.1. *There is a six-player exact game with a stable core that is not egalitarian stable. Hence core stability does not imply egalitarian stability, and by the preceding paragraph the threshold $n = 6$ is sharp.*

The witness is again a two-point generator game. Let $N = \{1, \dots, 6\}$,

$$A = (1, 1, 0, 1, 3, 0), \qquad B = (2, 0, 1, 0, 1, 2),$$

both with total 6, and $w_1(S) = \min\{A(S), B(S)\}$; the full coalition function is in the supplement. A second witness $w_2 = \min\{(3, 0, 2, 2, 1, 0), (1, 1, 1, 1, 2, 2)\}$ (total 8), found independently, confirms the phenomenon; all statements below are certified for both.

Proof of Theorem 7.1, non-ES half. By Lemma 2.1, w_1 is exact with $A, B \in \mathrm{Core}(w_1)$ and $w_1(N) = 6$; it is monotone. The procedure terminates in one round with

$$\widehat{A}^{w_1} = \big\{\{1,2\}, \{1,4\}, \{1,5\}, \{5,6\}, \{1,3,5\}, \{1,2,5,6\}, \{1,4,5,6\}\big\},$$

egalitarian claims $\widehat{\gamma} = \left(\frac{3}{2}, \frac{1}{2}, 1, \frac{1}{2}, \frac{3}{2}, \frac{3}{2}\right)$, and $\sum_i \widehat{\gamma}_i = \frac{13}{2} > 6 = w_1(N)$; by Lemma 3.1, w_1 is not egalitarian stable. (For w_2: claims $\left(\frac{3}{2}, 1, \frac{3}{2}, \frac{3}{2}, \frac{3}{2}, \frac{3}{2}\right)$, sum $\frac{17}{2} > 8$.) □

The stability half is where the new difficulty lies: external stability is a statement about *every* imputation outside the core, and no finite certificate for it exists in general. The witnesses, however, have the thinnest possible nondegenerate cores, and for such games we prove that stability *does* admit a finite certificate.

Segment cores. Exactly 12 coalitions S satisfy $A(S) = B(S) = w_1(S)$ (we call these *fully tight*); together with efficiency they force affine dimension 1, so $\mathrm{Core}(w_1) = \mathrm{conv}\{A, B\} = \{x_t := (1-t)A + tB \ : \ t \in [0,1]\}$, a segment (certificate in the supplement; likewise for w_2). Since $w_1 = \min\{A, B\}$ pointwise, every coalition is tight at A or at B.

Lemma 7.2 (domination from a segment)**.** *Let the core be the segment $[A, B]$ and let $S \subsetneq N$. The core elements attaining $v(S)$ are: the whole segment if S is fully tight; the endpoint A if $A(S) = v(S) < B(S)$; the endpoint B if $B(S) = v(S) < A(S)$. Consequently an imputation y is dominated via S only by a tight endpoint or, for fully tight S, by an interior point x_t.*

Proof. $x_t(S)$ is affine in t and $\ge v(S)$ on $[0,1]$, so its minimum over the segment is attained at an endpoint; it equals $v(S)$ exactly as listed. A dominating core element via S must satisfy $x(S) = v(S)$ (as $x(S) \le v(S)$ and $x \in$ Core), hence lies in the listed set. □

Lemma 7.3 (the domination boundary is cut by finitely many planes)**.** *For $S \subsetneq N$ define $\mathrm{opt}_S(y) = \max_{x \in F_S} \min_{i \in S}(x_i - y_i)$, where F_S is the set of core elements attaining $v(S)$, so that y is dominated via S iff $\mathrm{opt}_S(y) > 0$. Then opt_S is a concave, piecewise-affine function of y whose affine pieces have boundaries contained in the planes*

$$y_i = A_i, \qquad y_i = B_i, \qquad (y_i - A_i)(B_j - A_j) = (y_j - A_j)(B_i - A_i) \quad (i \neq j).$$

Proof. $\min_{i \in S}(x_i - y_i)$ is a minimum of affine functions of (t, y) and F_S is a segment or a point (Lemma 7.2), so opt_S is concave and piecewise affine. If F_S is an endpoint, the pieces of $\min_{i \in S}(x_i - y_i)$ change where two arguments tie or where the minimizing coordinate changes sign, i.e. across planes $y_i = A_i$ or $y_i = B_i$. If F_S is the whole segment, the interior maximizer over t is characterized by two coordinates $i, j \in S$ tying, $x_{t,i} - y_i = x_{t,j} - y_j$; eliminating t turns both the optimality boundary and the zero locus of the resulting affine piece into the stated pairwise plane for (i, j). A tie of three coordinates lies in the intersection of two pairwise planes, so pairwise planes suffice. □

Proposition 7.4 (finite stability certificate for segment cores)**.** *Let $\mathcal{P}$ consist of the efficiency plane, the coalition planes $y(S) = v(S)$, the floor planes $y_i = v(\{i\})$, and the planes of Lemma 7.3. If no vertex of the arrangement of $\mathcal{P}$ within the imputation set is an undominated imputation outside the core, then the core is stable.*

Proof. Let U be the set of undominated imputations, so $\mathrm{Core}(v) \subseteq U$ and stability is $U = \mathrm{Core}(v)$. By Lemma 7.3, $\{y : \mathrm{opt}_S(y) \leq 0\}$ is a sublevel set of a concave piecewise-affine function, i.e. a finite union of polyhedra delimited by planes of $\mathcal{P}$; intersecting over S and with the imputation set, U is a finite union of polytopes, each with all facets in $\mathcal{P}$. Suppose $U \supsetneq \mathrm{Core}(v)$ and pick a polytope piece $P_c \subseteq U$ with $P_c \not\subseteq \mathrm{Core}(v)$. The core is convex and cut out by inequalities $y(S) \geq v(S)$, so if every vertex of P_c lay in the core, then $P_c = \mathrm{conv}(\text{vertices}) \subseteq \mathrm{Core}(v)$; hence some vertex w of P_c — a vertex of the $\mathcal{P}$-arrangement — satisfies $w \in U \setminus \mathrm{Core}(v)$. □

Proof of Theorem 7.1, stability half (computer-assisted). For w_1 the family $\mathcal{P}$ has 89 planes. We enumerated all $\binom{89}{5} = 41{,}507{,}642$ potential vertex bases (with the efficiency plane always active); about 20.8 million are nonsingular, of which about 4.5 million solve to imputations outside the core. For *every* such vertex an exact rational dominating certificate was produced: a core element x^* with $x^*(S) = v(S)$ and $x^*_i > y_i$ for all $i \in S$, verified in exact arithmetic. No undominated vertex exists, so by Proposition 7.4 the core of w_1 is stable; the run for w_2 is identical in outcome. Two soundness guarantees accompany the computation. First, a vertex is discarded as dominated only on an exact rational witness as above; vertices with small floating-point domination margins are passed to a full exact undomination check, so no false negatives are possible. Second, the float classification of a vertex as an imputation outside the core is itself exact on this vertex set: by Cramer's rule every vertex coordinate is a rational with denominator bounded by the maximal basis determinant ($\leq 7\,000$ here by Hadamard's inequality), so every nonzero constraint value has magnitude at least 1.4×10^{-4}, far exceeding both the classification tolerance (10^{-7}) and the solve error ($\approx 10^{-12}$). The enumeration architecture was gated before use: it re-finds the known escapes of two non-stable segment-core control games (one five-player, one six-player), including an escape lying on the new plane classes. Full run statistics, checkpoints, and code are in the supplement. □

Corollary 7.5 (full resolution)**.** *Neither exactness nor core stability implies egalitarian stability. The thresholds are sharp: exact games are egalitarian stable precisely up to $n = 4$ (Corollary 5.5), and stable games precisely up to $n = 5$.*

Remark 7.6 (why the stable half is harder, and where the witnesses live)**.** *Non-ES requires inflated claims, which by Remark 6.3 concentrate on thin cores; but thin cores are generically* not *stable — in our experiments, escape imputations were certified for every one of* 146 *randomly generated non-ES games at $n \in \{5, 6\}$. The witnesses sit at the extreme point of this trade-off: a one-dimensional core that nevertheless dominates every outside imputation. This coupling — thinness driving non-ES while destroying stability — is also why the stable half could not be settled by sampling, and why the witnesses had to be certified rather than searched. Both witnesses are exact but not large-core (automatic, since large-core games are ES), so they simultaneously exhibit the exact/ES and the stable/large gaps at $n = 6$.*

8 Verification methodology

All numerical claims are certified in exact rational arithmetic. The egalitarian procedure was implemented four times, independently, by different authorship processes; each implementation was gated on exact reproduction of Examples 2 and 3 of Dietzenbacher (2021) (claims, admissible collections, strong claimants, and solution values) before any search was run, and one implementation was additionally calibrated on 235 random exact four-player games, all of which are egalitarian stable as required by the results of Biswas et al. (1999) and Dietzenbacher (2021). The three witnesses reported here were each discovered independently and cross-verified through all implementations; exactness is certified by exhibiting, for every proper coalition, a core allocation attaining its worth (linear programs solved in exact rational arithmetic by a solver fuzz-tested against a standard floating-point implementation). The stability certificate of Theorem 7.1 was discharged by exhaustive enumeration of the $\binom{89}{5}$ vertex bases of the certifying arrangement for each six-player witness; every outside-core vertex was discarded only against an exact rational dominating core element, borderline cases were passed to a full exact undomination check, and the enumeration architecture was validated by re-finding the known escapes of two non-stable segment-core control games before use. The six-player witnesses' exactness, non–egalitarian-stability, and segment-core geometry were additionally re-derived by a second, independently written toolchain (separate exact LP solver and separate implementation of the egalitarian procedure). A machine-readable supplement contains the full certificate data — coalition functions, procedure traces, claims, admissible collections, solution values, core violations, and per-coalition exactness witnesses — together with a self-contained verifier using only the Python standard library. The computational search, the verification tooling, and the drafting of this paper made substantial use of large-language-model assistance; every mathematical claim was verified by the certification pipeline described here, and all text was reviewed by the author.

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